

Worksheet 2

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Subject Name: Backend Technologies

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Aim/Overview of the practical:

To develop a Signup Page using NodeJS and store the user-entered data into a file.

Objectives:

- To design a Signup and Login page using HTML and Node.js.
- To create an HTTP server using Node.js.
- To handle HTTP GET and POST requests.
- To validate username, email, password, and confirm password.
- To ensure that the username starts with a capital letter.
- To verify the user's email using a 6-digit OTP.
- To implement OTP expiration for 5 minutes.
- To prevent duplicate usernames and email addresses.
- To store registered user data in a text file.
- To hash passwords before storing them.
- To authenticate users during login.
- To understand the use of Node.js core modules such as http, fs, querystring, and crypto.

Software Require:

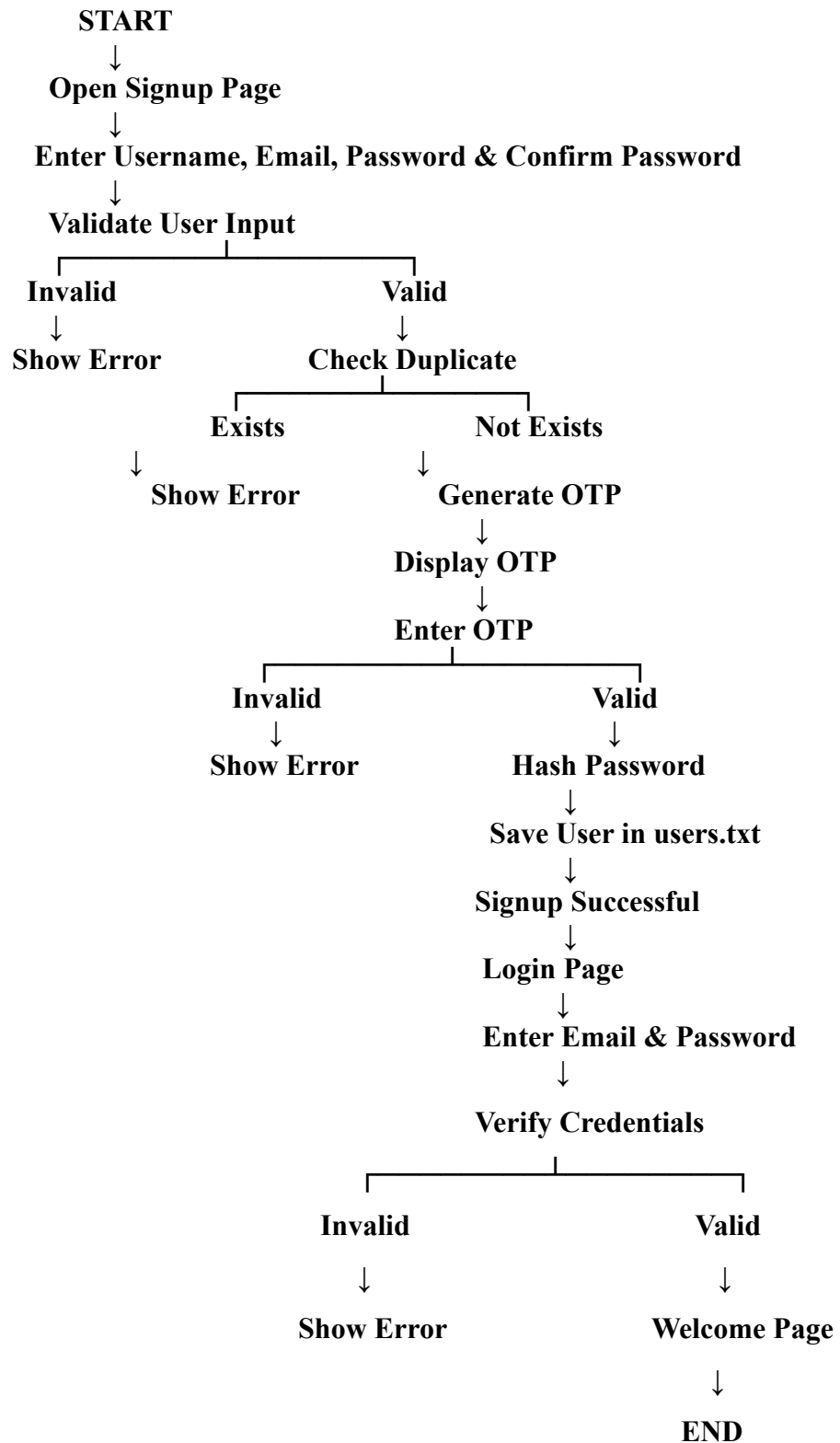
- NodeJS Installed
- VS Code
- Web Browser
- Terminal

Project Structure:

SignupProject/

```
|
|
|— server.js
|
|— users.txt
```

FlowChart:



Code/Output:

Import Required Modules

```
const http = require("http");  
const fs = require("fs");  
const querystring = require("querystring");  
const crypto = require("crypto");
```

Create HTTP Server

```
const server = http.createServer((req, res) => {  
  const url = req.url.split("?")[0];  
  // Request handling code  
});
```

Signup Page

```
<form method="POST" action="/" autocomplete="off">  
<input  
  type="text"  
  name="username"  
  placeholder="Example: Saloni"  
  required>  
  
<input  
  type="email"  
  name="email"  
  placeholder="Enter Email"  
  required>  
  
<input  
  type="password"  
  name="password"  
  placeholder="Enter Password"  
  required>  
  
<input  
  type="password"  
  name="confirmPassword"  
  placeholder="Confirm Password"  
  required>  
  
<input  
  type="checkbox"  
  name="terms"  
  required>  
  
<button type="submit">  
  Sign Up  
</button>  
  
</form>
```

Username Validation

```
const usernamePattern =  
  /^[A-Z][A-Za-z0-9_]{2,19}$/;  
  
if (!usernamePattern.test(username)) {  
  
  // Show invalid username message  
  
  return;  
}
```

Email Validation

```
const emailPattern =  
  /^[^\\s@]+@[^\\s@]+\\.\\.[^\\s@]+$/;  
  
if (!emailPattern.test(email)) {  
  
  // Show invalid email message  
  
  return;  
}
```

Password Validation

```
const passwordPattern =  
  /^(?=.*[a-z])(?=.*[A-Z])(?=.*\\d)(?=.*[@$!%*?&]).{8,}$/;  
  
if (!passwordPattern.test(password)) {  
  
  // Show password error  
  return;  
}
```

Confirm Password

```
if (password !== confirmPassword) {  
  
  // Show password mismatch  
  return;  
}
```

Generate OTP

```
const otp =  
  crypto.randomInt(  
    100000,  
    1000000  
  ).toString();
```

Store User Data

```
const user = `  
  Username : ${storedData.username}  
  Email : ${email}  
  Password Hash : ${hashedPassword}  
  Email Verified : YES  
  Registration Date : ${new Date().toLocaleString()}  
`;  
`;
```

```
fs.appendFile(  
  "users.txt",  
  user,  
  (err) => {  
    // Handle result  
  }  
);
```

Login Page

```
<form method="POST" action="/login">
```

```
<label>Email</label>
```

```
<input  
  type="email"  
  name="email"  
  required>
```

```
<label>Password</label>
```

```
<input  
  type="password"  
  name="password"  
  required>
```

```
<button type="submit">  
  Login  
</button>  
</form>
```

Login Authentication

```
const hashedPassword =  
  crypto  
    .createHash("sha256")  
    .update(password)  
    .digest("hex");
```

```
fs.readFile(  
  "users.txt",  
  "utf8",  
  (err, fileData) => {  
  
    // Search for matching  
    // email and password hash  
  
  }  
);
```

Output:

Create Account

Username

Saloni1

Must start with a capital letter. 3-20 characters allowed.

Email

saloniraj@gmail.com

Password

.....

Show

Minimum 8 characters, uppercase, lowercase, number and special character.

Confirm Password

.....

Show

☒ I agree to the Terms & Conditions

Sign Up

Already have an account? [Login](#)



Login

Email

saloniraj8757@gmail.com

Password

.....

Login

Don't have an account? [Sign Up](#)

Email Verification

Verification code for:

saloniraj@gmail.com

For this practical project, the OTP is displayed below.

1 8 6 6 4 3

Enter 6-digit OTP

Verify Email

OTP expires in 5 minutes.



Welcome, Saloni!

☒ Login Successful

You have successfully logged into your account.

Email: **saloniraj8757@gmail.com**

Logout



Signup Successful!

Your email has been verified.

Your account has been created successfully.

Username: **Saloni1**

[Go to Login](#)

```
SignupProject > users.txt
1
2
3 Username : Saloni
4 Email : saloniraj8757@gmail.com
5 Password Hash : decaee5c012d0247b0ded62b52197038
6 Email Verified : YES
7 Registration Date : 8/8/2026, 11:00:33 pm
8 -----
9
10
11
12 Username : Saloni1
13 Email : saloniraj@gmail.com
14 Password Hash : decaee5c012d0247b0ded62b52197038
15 Email Verified : YES
16 Registration Date : 8/8/2026, 11:32:16 pm
17 -----
```

Learning Outcomes:

- Understand the basics of backend development using Node.js.
- Create Signup and Login pages using HTML and Node.js.
- Handle HTTP GET and POST requests.
- Read and process user input from web forms.
- Generate and verify OTPs.
- Implement email-verification logic using OTP.
- Validate usernames, emails, and passwords.
- Hash passwords before storing them.
- Store and retrieve user data using the Node.js File System module.